

Optimization Follow-Up: cVAE Training Tricks Without Architecture Changes

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1 Goal

This short follow-up isolates a narrow question: can we improve the current cVAE by changing only the optimization recipe, while leaving the architecture unchanged?

The reference model is the best run from the main report, `cvae_nb_005`, with:

- latent size 64,
- learning rate $3 \cdot 10^{-4}$,
- batch size 64,
- $\beta = 10^{-3}$,
- the same UrbanSound8K spectrogram setup as in the main report.

2 What Changed

We tested a sequence of optimization-only follow-ups while keeping the architecture and data preprocessing fixed.

Follow-up A: warmup + free bits. This run added:

- linear KL warmup over the first 25 epochs;
- free bits with threshold 0.01 per latent dimension.

Run id: `cvae_tricks_20260411_210250_lat64_lr0.0003_bs64_bk10.001_warm25_fb001`.

Follow-up B: EMA on top of the same schedule. We then added exponential moving average weights with `ema_decay=0.999`, while leaving the previous warmup/free-bits settings unchanged. Run id: `cvae_ema_20260412_181315_lat64_lr0.0003_bs64_bk10.001`.

Follow-up C: EMA + retuned schedule. Finally, we kept EMA but shortened and softened the KL regularization schedule:

- KL warmup reduced to 10 epochs;
- free bits reduced to 0.005;
- training length increased to 120 epochs;
- patience increased to 30.

Run id: `cvae_ema_tricks_20260412_183413_lat64_lr0.0003_bs64_bk10.001_warm10_fb0005`.

3 Results

Table 1 compares the previous best cVAE with all three follow-up runs.

Table 1: Best baseline cVAE vs optimization-only follow-up series.

Run	Best val	Log-mel L1	MR-STFT	FAD	Div. ratio
<code>cvae_nb_005</code>	0.1058	0.0918	1.8971	3.0198	0.5711
<code>cvae_tricks_20260411_210250</code>	0.1106	0.0985	1.9268	16.0544	0.5136
<code>cvae_ema_20260412_181315</code>	0.4291	—	—	—	—
<code>cvae_ema_tricks_20260412_183413</code>	0.1048	0.0909	1.8873	3.2587	0.5588

The results separate into three clear regimes:

- warmup + free bits alone hurt both reconstruction and generation badly;
- adding EMA on top of that untuned schedule was unstable and collapsed validation quality very early;
- EMA combined with a shorter warmup, smaller free-bits threshold, and longer training produced the best validation objective and the best reconstruction metrics of all runs.

However, the new best training objective does *not* translate into the best generation score. The strongest EMA-based run improves over `cvae_nb_005` on best validation loss, log-mel L1, and MR-STFT, but its FAD is slightly worse (3.2587 vs 3.0198), and its diversity ratio is also slightly lower (0.5588 vs 0.5711).

So the updated conclusion is more nuanced than in the first follow-up: EMA plus a better-tuned KL schedule improves reconstruction, but the original `cvae_nb_005` is still slightly better as a generator under the current FAD metric.

4 Interpretation

This follow-up still narrows the search space in a useful way. Two conclusions are now fairly clear.

- EMA is not automatically helpful in this setup; with the original warmup/free-bits schedule it was clearly harmful.
- EMA becomes useful only after the KL schedule is retuned. In that regime it improves reconstruction, but does not yet beat the original best checkpoint on prior-matching quality.

One plausible interpretation is that the retuned EMA run learns a cleaner reconstruction manifold, while the older `cvae_nb_005` still matches the sampling prior slightly better under standard-normal generation. That would explain why the new run wins on reconstruction metrics but loses on FAD.

5 Artifacts

The main artifacts for this follow-up series are stored under:

- `experiments/cvae/cvae_tricks_20260411_210250_lat64_lr0.0003_bs64_bk10.001_warm25_fb001/`
- `experiments/cvae/cvae_ema_20260412_181315_lat64_lr0.0003_bs64_bk10.001/`
- `experiments/cvae/cvae_ema_tricks_20260412_183413_lat64_lr0.0003_bs64_bk10.001_warm10_fb0005/`
- each run stores `train_summary.json`;
- the evaluated runs also store `eval/metrics.json` and `eval/qualitative/`.

Conclusion

The optimization-only follow-up series produced one meaningful improvement: EMA with a retuned KL schedule gives the best validation objective and the best reconstruction metrics seen so far. But it still does not beat the older `cvae_nb_005` on FAD, so the best generator in the current report remains unchanged. The practical takeaway is that optimization tricks can help, but they are shifting the reconstruction-generation trade-off rather than solving it outright.